

Know.	App.	Think.	Comm.
/23	/17	/5	/3

Knowledge and Understanding [23 marks]

1. Multiple Choice: Write the CAPITAL LETTER corresponding

to the correct answer on the line provided. [1 mark each]

i. The average rate of change of $f(x) = x^2 + 3x + 4$ over the interval $0 \leq x \leq a$ is:

- (A) $3a$ (B) $a^2 + 3$ (C) $a + 3$ (D) none of these

ii. The function $f(x)$ has a jump discontinuity at $x=a$ if

- (A) $\lim_{x \rightarrow a} f(x)$ exists (B) $f(a)$ is undefined
 (C) $\lim_{x \rightarrow a} f(x) = f(a)$ (D) $\lim_{x \rightarrow a} f(x) \neq \lim_{x \rightarrow a^+} f(x)$

iii. Evaluate the limit, if it exists: $\lim_{x \rightarrow \infty} \frac{(3x^2 - 1)^2}{2x^4 - 3x^2 + 1}$

- (A) $\frac{3}{2}$ (B) ∞ (C) 0 (D) $\frac{9}{2}$

iv. If $f(x) = x^2 - 1$, the value of $\left. \frac{dy}{dx} \right|_{x=2}$ is

- (A) 2 (B) 4 (C) 0 (D) 1

2. Given the function $y = f(x)$, determine the following: [1 mark each]

~~a)~~ $\lim_{x \rightarrow 5^-} f(x) = \text{omit}$

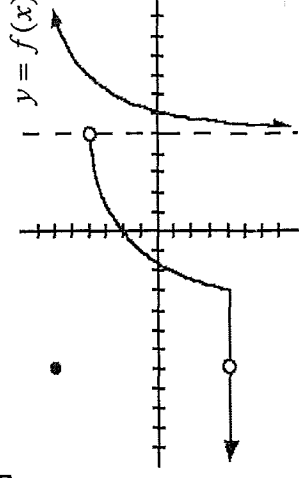
d) $f(5) =$

b) $\lim_{x \rightarrow 5^+} f(x) =$

e) $f(-7) =$

c) $\lim_{x \rightarrow 5} f(x) =$

f) $\lim_{x \rightarrow -7} f(x) =$



3. Evaluate the following limits. [10 marks]

a) $\lim_{x \rightarrow 1} \frac{1}{\sqrt{x} - 1} = 1$ ✓✓✓

b) $\lim_{x \rightarrow 1} \frac{x^3 - 5x^2 - x + 5}{x^3 + 2x^2 - x - 2} = 1$ ✓✓✓

c) $\lim_{x \rightarrow 1} \frac{\sqrt[4]{x} - 1}{\sqrt[8]{x} - 1} = 1$ ✓✓✓✓

4. Sketch a single function $y = f(x)$ that has all of the following properties: ✓✓✓

i. $\lim_{x \rightarrow 2^+} f(x) = -\infty$

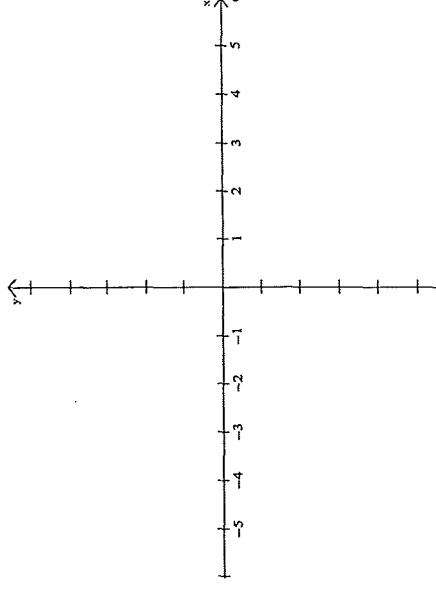
iv. $\lim_{x \rightarrow 1^+} f(x) = 2$

ii. $\lim_{x \rightarrow 2^-} f(x) = \infty$

v. $\lim_{x \rightarrow 1^-} f(x) = 2$

iii. $\lim_{x \rightarrow -\infty} f(x) = 0$

vi. $f(1) = -2$



Application [17 marks]

1. The position of a particle moving along the x-axis is given by $s(t) = t^2 - 5t + 4$ where t is the elapsed time in seconds.

(a) Calculate the average velocity of the particle from $t = 2s$ to $t = 5s$. ✓✓

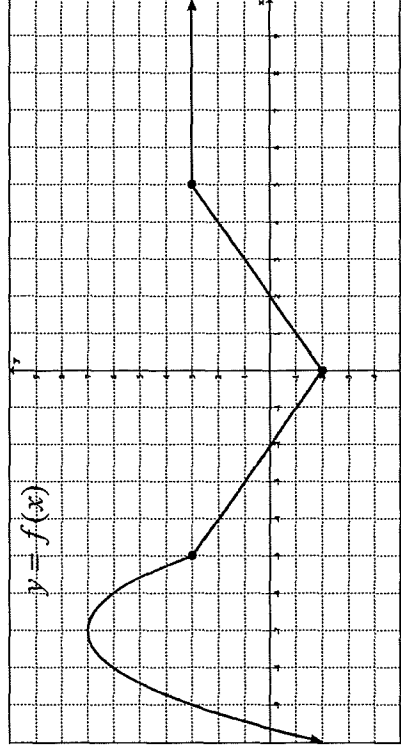
(b) Calculate the instantaneous velocity of the particle when $t = 2s$. ✓✓✓

2. A tangent to the parabola $y = \sqrt{ax+b}$ is perpendicular to $3x + y + 4 = 0$ at point $(4,3)$. Determine the values of a and b . ✓✓✓✓✓

3. Function $g(x) = \frac{x-3}{\sqrt{5x+1} - \sqrt{3x+7}}$ is given. For what value of the constant k is the function

$$f(x) = \begin{cases} g(x) & x \in D_g \\ k & x = 3 \end{cases} \text{ continuous on its domain? } \checkmark\checkmark\checkmark\checkmark$$

4. Graph the derivative function on the grid for the function $y = f(x)$ given below. ✓✓✓



Thinking [5 marks]

1. Given the function $f(x) = \frac{x^3 + ax^2 + bx + 5}{x^3 + 2x^2 - x - 2}$, determine whether there are numbers a and b such that both $\lim_{x \rightarrow 1} f(x)$ exists, and $\lim_{x \rightarrow -1} f(x)$ exists. If so, evaluate both limits. ✓✓✓✓✓

Communication [3 marks]

Determine at which value(s) of x the following function is not differentiable. Explain your reasoning clearly. [3 marks]

